

LOGISTIC REGRESSION

- Logistic Regression is a supervised machine learning algorithm used to predict a binary outcome (like yes or no) by calculating the probability of an event. The primary goal of Logistic Regression is to model a relationship between a set of independent variables (features) and a binary dependent variable (target) in such a way that it can predict the probability of the binary outcome (e.g. predicting if an email is spam (1) or not spam (0), a tumor is malignant (1) or benign (0), loan approved (1) or not approved (0) etc.). It solves classification problems. It outputs a probability value between 0 and 1.
- **Types**
 1. **Binomial Logistic Regression:** Used when the dependent variable has only two possible categories, such as Yes/No, Pass/Fail, or 0/1. It is the most common type and is used for binary classification tasks.
 2. **Multinomial Logistic Regression:** Used when the dependent variable has three or more unordered categories. For example, classifying animals as cat, dog, or sheep. It extends logistic regression to handle multiple classes.
 3. **Ordinal Logistic Regression:** Used when the dependent variable has three or more categories with a natural order, such as Low, Medium, and High. It considers the ranking of categories during prediction.
- **Financial Applications**
 - ❖ **Credit Scoring & Risk Assessment:** Estimating the Probability of Default (PD) for loan or bond applicants.
 - ❖ **Fraud Detection:** Flagging suspicious credit card or banking transactions.
 - ❖ **Corporate Bankruptcy Prediction:** Classifying firms as financially stable or distressed ("good" vs. "bad").
 - ❖ **Loan Approval Automation:** Deciding whether to approve or reject retail credit line
- **Assumptions**
 1. **Independent Observations:** Each data point should be independent of the others, meaning there should be no dependence between observations.
 3. **Binary Dependent Variable:** The target variable is typically binary and can take only two values, such as 0/1 or Yes/No.
 4. **Linearity of Log-Odds:** The independent variables should have a linear relationship with the log-odds of the dependent variable.
 4. **No Extreme Outliers:** Extreme outliers can distort coefficient estimates and negatively affect model performance.
 5. **Large Sample Size:** A sufficiently large dataset helps produce more reliable and stable predictions.

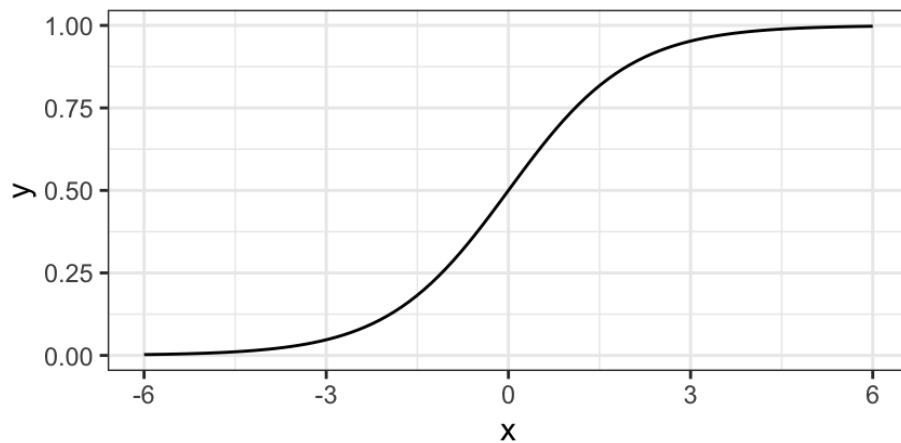
➤ **Terminologies Used**

1. **Independent Variables:** These are the input features or predictor variables used to make predictions about the dependent variable.
2. **Dependent Variable:** This is the target variable that we aim to predict. In logistic regression, the dependent variable is categorical.
3. **Logistic Function:** This function transforms the independent variables into a probability between 0 and 1 which represents the likelihood that the dependent variable is either 0 or 1.
4. **Odds:** This is the ratio of the probability of an event happening to the probability of it not happening. It differs from probability because probability is the ratio of occurrences to total possibilities.
5. **Log-Odds (Logit):** The natural logarithm of the odds. In logistic regression, the log-odds are modelled as a linear combination of the independent variables and the intercept.
6. **Coefficient:** These are the parameters estimated by the logistic regression model which shows how strongly the independent variables affect the dependent variable.
7. **Intercept:** The constant term in the logistic regression model which represents the log-odds when all independent variables are equal to zero.
8. **Maximum Likelihood Estimation (MLE):** This method is used to estimate the coefficients of the logistic regression model by maximizing the likelihood of observing the given data.

➤ Logistic regression is a type of linear model that falls under the generalized linear models (GLM) family.

Suppose we have the categorical dependent variable Y (loan approved (1) or not approved (0)) and an independent variable X (savings). If we use the linear function $Y = \beta_0 + \beta_1 X + \epsilon$ then as the values of X changes, Y can take on a value from $(-\infty, \infty)$. But probabilities are confined to $[0,1]$. Using this principle of linear model, we cannot directly model the probabilities for a binary outcome. Instead, we need a logistic model to make sense of the probabilities. Therefore, we want to apply a transformation to the input so the outcome can be confined. This transformation is known as the logistic regression equation.

- Logistic regression fits an S-shaped curve to map input values to a probability. The data points are assigned to one of the classes by estimating the probabilities. To estimate the probabilities, logistic regression uses **Logistic Function (Sigmoid Function)**. **The sigmoid function maps the linear combination of input features to a value between 0 and 1.** This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the **sigmoid curve or logistic curve**. We can use 0.5 as the threshold.



Consider an input feature represented as a matrix $X = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}$ and the dependent variable Y having binary values 0 or 1.

In the linear regression model, we model the relationship between outcome and features with a linear equation: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$

For classification, we prefer probabilities between 0 and 1, so we change the right side of the equation into the logistic function. This makes the output to assume only values between 0 and 1.

$$P(Y = 1) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}}$$

➤ ODDS / ODDS RATIO

The interpretation of the weights in logistic regression differs from the interpretation of the weights in linear regression since the outcome in logistic regression is a value between 0 and 1. To interpret the weights, we need to reformulate the equation for the interpretation so that only the linear term is on the right side of the formula.

$$\ln\left(\frac{P(Y = 1)}{1 - P(Y = 1)}\right) = \ln\left(\frac{P(Y = 1)}{P(Y = 0)}\right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

The term $\frac{P(Y=1)}{1-P(Y=1)}$ is called “**odds**” (*probability of event divided by probability of no event*), and $\ln\left(\frac{P(Y=1)}{P(Y=0)}\right)$ is called **log odds**. This formula shows that the **logistic regression model is a linear model for the log odds**. This transformation allows logistic regression to output valid probabilities,

Let us apply *exp* function to both sides of the above equation:

$$\frac{P(Y = 1)}{1 - P(Y = 1)} = odds = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}$$

Consider an example: Suppose a basket contains 3 apples and 5 oranges.

- The probability of picking an orange is $5/(3+5) = 0.625$

- The odds of picking an orange are $5/3 \approx 1.667$

This means that picking an orange is ≈ 1.667 times more likely than picking an apple. Conversely, the odds of picking an apple are $3/5 = 0.6$, which is less than 1.

- The odds ratio helps to interpret the effect of model coefficients. The odds ratio quantifies the relationship between a binary outcome and a predictor variable.
- The odds ratio is used to understand how a one-unit change in a predictor variable affects the odds of an event occurring. It helps to assess the strength and direction of relationship between the predictor and the binary outcome.
- It is calculated as the ratio of odds of an event occurring in one group to the odds of the same event occurring in another group.

$$\frac{Odds_{xj+1}}{Odds_{xj}} = \exp(\beta_j)$$

- A change in a feature by one unit, changes the odds ratio (multiplicative) by a factor of $\exp(\beta_j)$. This multiplier is the odds ratio. A change in x_j by one unit increases the log odds ratio by the value of the corresponding weight.
 - The odds ratio gives logistic regression its interpretability. The magnitude of the coefficient reflects how strong this influence is, and the odds ratio (which is the exponential of the coefficient) tells us how much the odds change for a one-unit increase in the variable.
 - It tells how the odds of an event change based on inputs, which is useful in many applied settings like healthcare, marketing and finance.

For example, if the odds is 2, it means that the probability for $Y = 1$ is twice as high as $Y = 0$. If we have a weight (log odds ratio) of 0.7, then increasing the respective feature by one unit multiplies the odds by $\exp(0.7)$ (approximately 2). The odds change to 4. But usually you do not deal with the odds and interpret the weights only as the odds

ratios. Because for actually calculating the odds, you would need to set a value for each feature, which only makes sense if you want to look at one specific instance of your dataset.

➤ More specifically, in logistic regression:

- A positive coefficient means the log-odds of the outcome increase as the input increases. This corresponds to an increase in probability.
- A negative coefficient means the log-odds decrease as the input increases. This corresponds to a decrease in probability.
- A coefficient of zero means that the variable has no effect on the outcome.

For example, if we want to predict whether a loan is approved or not approved ($Y=1$ for approved, $Y=0$ for not approved) based on whether they still have an existing debt. Let $X_1=0$ mean that they have no existing debt and $X_1=1$ mean that they have existing debt. The log-odds of $Y = \text{approval}$ will be $\beta_0 + \beta_1 X_1$.

The coefficient β_1 , then represents the change in log-odds of being approved when the person has an existing debt, compared to someone who does not. To make this more interpretable, we can exponentiate β_1 to get the odds ratio:

- If β_1 is positive, $\exp(\beta_1)$ is greater than 1, meaning having existing debt increases the odds of being approved.
- If β_1 is negative, $\exp(\beta_1)$ is less than 1, meaning having existing debt decreases the odds of approval.
- If β_1 is 0, $\exp(\beta_1)$ is 1, meaning debt status has no effect.

➤ **Steps:**

- **Linear step:** It starts like linear regression by computing a linear combination of input features.
- **S-curve transformation:** It passes that linear score through a sigmoid function (also called the logistic function).
- **Probability mapping:** The sigmoid function squashes any real-valued number into a smooth S-shaped curve ranging from 0 to 1.
- **Decision boundary:** A threshold (usually 0.5) decides the final class label; values above 0.5 map to 1, and values below map to 0

➤ **Evaluation Metrics for Logistic Regression**

- *Accuracy*
- *Precision*
- *Recall*
- *F1 Score*
- *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*

- **Advantages**

- ❖ Logistic regression model is not only a classification model, but also gives the probabilities. This is a big advantage over models that can only provide the final classification.
- ❖ Logistic regression can also be extended from binary classification to multi-class classification.

- **Limitations**

- ❖ Logistic regression has been widely used by many different people, but it struggles with its restrictive expressiveness (e.g., interactions must be added manually), and other models may have better predictive performance.
- ❖ The interpretation is more difficult because the interpretation of the weights is multiplicative and not additive.
- ❖ If there is a feature that would perfectly separate the two classes, the logistic regression model can no longer be trained. This is because the weight for that feature would not converge, because the optimal weight would be infinite. This is really a bit unfortunate because such a feature is really useful. But you do not need machine learning if you have a simple rule that separates both classes. The problem of complete separation can be solved by introducing penalization of the weights or defining a prior probability distribution of weights.